SILTHAUL · DESIGN DECISIONS REGISTER

SiltHaul design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-11-03	Amish Chadha	Kit 1.9.1: concept selection opened for Amish (SLH-DDR-005, Proposed, awaiting Amish); STEM fixes recorded as decided under the standing instruction (SLH-DDR-006)
0.3	2026-10-03	Amish Chadha	Open decision 1 decided (round 2, 2A) and recorded as SLH-DDR-004; no open decisions
0.2	2026-10-03	Amish Chadha	Amish's requirement decisions 2A and 7A recorded (SLH-DDR-003); boot reading for R9 opened for Amish
0.1	2026-10-03	Amish Chadha	Register opened; every decision made under Amish's pre-approval of 2026-10-03

SiltHaul design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list decisions.

> **Safety:** Several decisions below set SiltHaul's safety case (overload limit, holding, anchors, dumping). Each took the conservative option; the evidence that would relax it is in SLH-DDR-001, Table 1.

Open decisions

One. Open decision 1 (what "fits a small car boot" means) was decided by Amish on 2026-10-03; see Decisions made and SLH-DDR-004.

#	Decision needed	Options	Recommendation	Affects in the build	Source
2	Which concept option to carry forward	A hand-capstan loop (126 of 160); B crew-hauled box (133); C hand slurry pump (96); or a hybrid	Proposed, awaiting Amish: no pick, because A and B are within 10 % (5.3 %); C is not recommended	Option A is the drawn, priced and planned design; choosing B means redoing the BOM, drawings and build plan	SLH-DDR-005, SLH-OPT-001

To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Release load of the shear pins from the bar actually bought	The 4 mm pin is sized on 240 MPa shear; a stronger bar raises the overload limit above 3,000 N	SLH-CAL-001, F; R11
2	Recommended loads of the M12 wedge anchor bought, in the concrete grade expected	The tail block check uses 4 kN tension and 6 kN shear	SLH-CAL-001, I; R7
3	Rope breaking strength and splice efficiency of the rope bought	Factor 5.4 at the limit assumes 18 kN and 90 % at the splice	SLH-CAL-001, G; R6
4	Sheave bore and rating	Pins are 25 mm; the sheaves must be rated at least 1,000 kg	BOM line 14
5	Bearing centre heights and bolt pitches	The pads and top plates are drilled to UCP206 and UCP205 catalogue sizes	SLH-DWG-101
6	Sprocket bores and the 12-tooth sprocket's fit on the 36 mm hub	Sets the hub and the line-up with the 48-tooth wheel	SLH-DWG-104

Value engineering

Value-engineering target: USD 2,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 1,055.60 (USD 944.40 under the target). Main cost drivers and savings worth trying:

- The two boxes (USD 160) and the winding drum (USD 98) are the largest made parts; the rope (USD 80.60), the ramp (USD 73) and the frame (USD 72) follow.
- Bought lifting gear (sling, shackles, two bridle sets, USD 164) is kept at rated, traceable grades because it carries the safety case.
- Savings worth trying: a drum from a scrap gas cylinder or pipe offcut of the same diameter; a timber ramp with a steel wear strip; buying rope with factory eye splices in bulk.

Decisions made

Date	Decision	Decided by	Record
2026-10-03	Split winding drum in place of a friction capstan; 4:1 chain drive with two cranks	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, items 1, 2
2026-10-03	Shear pin limits rope tension to about 2.5 kN; all parts sized on 3,000 N; requirement R11 added	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 3
2026-10-03	Two opposite ratchet wheels and pawls on the drum shaft; requirement R12 added	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 4
2026-10-03	Tail block on four drilled floor anchors only, never walls or door frames	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 5
2026-10-03	Capstan anchored by a round sling to a tree or vehicle, level or rising 10 degrees at most; stakes only against skating	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 6
2026-10-03	10 mm polyester double braid, 18 kN	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 7
2026-10-03	Box open toward the door, sloped back, forward tipping; two people on the tipping bar	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, items 8, 9
2026-10-03	Capstan at least 6 m beyond the door	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 10
2026-10-03	First co-design candidate to approach: Kerala State Disaster Management Authority (not agreed)	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 11
2026-10-03	SiltHaul capstan recorded as the candidate common block for SaltDrag; CalRig as the first candidate proof-load rig	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 12
2026-10-03	`budget_usd` kept at 2,000 as a value-engineering target	Amish: "I also accept any cost overruns or variations from the assumed scope cost."	SLH-DDR-001, item 13

Date	Decision	Decided by	Record
2026-10-03	R1 kept as the trial target and reported not met on paper (superseded by 2A below)	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, Consequences
2026-10-03	Design for construction: the thirteen changes of SLH-DDR-002	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-002
2026-10-03	Appearance model additions for renders: wound rope on the drum, stakes above ground only, context wall, tree and mannequin	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	docs/ REVIEW.md, TRL 3
2026-10-03	2A: second box on the return leg so the rope loop carries mud both ways; R1 restated as "no lifting or carrying of mud; output at least 0.5 times the same crew with shovels and buckets" (now 0.57, met on paper)	Amish: "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"	SLH-DDR-003, items 1 to 5
2026-10-03	7A: bolted flat-pack frame (two welded side frames, four bolted cross members) that fits a small car boot, keeping the 25 kg heaviest-part limit (heaviest lift 23.1 kg)	Amish: "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"	SLH-DDR-003, items 6 to 8
2026-10-03	2A (round 2): R9 accepted with the rear seats folded and restated as "fits a small hatchback with the rear seats folded; heaviest part 25 kg or less"; met on paper (heaviest lift 23.1 kg; kit about 454 L in 594 L). No design change	Amish: "i agree with all the 46 recommendations you provided. please proceed."	SLH-DDR-004
2026-11-03	Foot boards with heel cleats and stakes added to option B; diaphragm stroke of option C cut from 120 to 40 mm (STEM fixes)	Decided under Amish's standing instruction of 2026-10-23 (fix keeps the working principle)	SLH-DDR-006